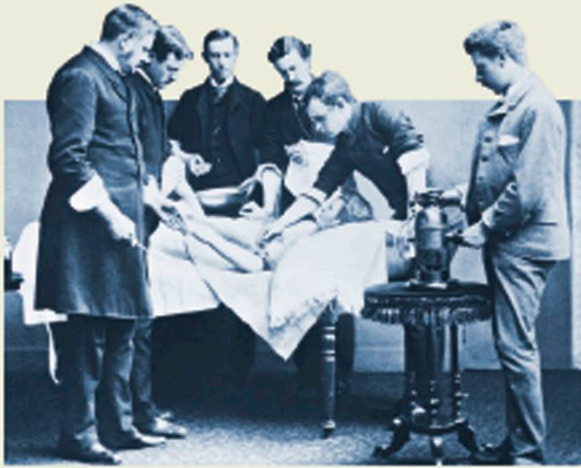




An antiseptic spray is used during an operation, c 1870

Hospital care

In the past, hospitals were dirty, overcrowded places. Two of the people who helped to change all that were British surgeon Joseph Lister (see p.144), who introduced antiseptics, making surgery safer, and British nursing pioneer Florence Nightingale, who demonstrated that clean hospitals prevented infection and helped sick people get better faster.

“The very first requirement in a hospital is that it should do the sick no harm.”

Florence Nightingale, *Notes on Nursing*, 1860

Medical aids

Tools such as tweezers and scalpels (sharp-bladed knives), similar to those doctors use today, date back to ancient times. Over the centuries, as medical knowledge advanced, the need arose for more complicated instruments, such as endoscopes and blood pressure monitors to identify and treat patients' symptoms.



Wood and pewter syringe, late 1700s

The tube ends in a rubber bulb, which is placed on the skin and inflated.

The funnel directs light from the candle.

Viewing lens

Nozzle



Otoscope for looking inside the ear, 1890s

Deadly diseases

In most parts of the world, people live far longer than they did 500 years ago, when average life expectancy was about 40 years. With vaccines and antibiotics, medical workers can now prevent or cure many infectious diseases that would previously have killed thousands.

+ Smallpox

Once a feared killer, smallpox has now been wiped out thanks to a global immunization campaign.

+ Polio

This highly infectious viral disease targets children. Vaccines are helping to end it worldwide.

+ Plague

Bubonic plague (the Black Death) killed millions of people in Europe in the 1300s. Outbreaks still occur today, but the disease is treatable with antibiotics.



Engraving of a doctor wearing a beak-shaped mask for protection against plague, 1656

1865

Joseph Lister introduced life-saving standards of cleanliness and hygiene into operating rooms and hospitals.

1895

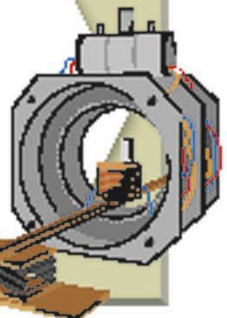
German physicist Wilhelm C. Röntgen produced the first X-ray photograph. X-rays, which can penetrate skin and tissue to reveal bone, have become a key tool in medicine.

1928

Sir Alexander Fleming, a Scottish biologist, discovered penicillin, heralding the era of antibiotics. Penicillin was first used as an effective drug in the 1940s.

1977

The first MRI (magnetic resonance imaging) scan was carried out. MRI creates multiple images of the body without exposing it to damaging radiation.



First MRI machine